

of social signals is not computationally tractable. We are working on a future model that uses different learning dynamics and we hope will resolve this issue of computational tractability. For now we have minimal simulation results to report (see Appendix B).

## Appendix B Reinforcement Learning Dynamics

It is worth noting that nothing about the Generalized Bach or Stravinsky game requires agents to learn according to replicator dynamics. In this Appendix, we give some limited simulation results for the model in which the discrete replicator dynamics is replaced by agents learning according to Roth-Erev reinforcement learning with forgetting (Erev and Roth, 1998). A key benefit of reinforcement learning for future research is that it does not require quantifying the utility of an individual’s strategy profile relative to the utility of all other profiles present in the subset of the population that shares that individual’s coordination preferences. So, if one wanted to model a population whose preferences were evolving dynamically, the learning dynamics would function normally even if the dynamics lead to every individual in the population having unique coordination preferences.

Our implementation of reinforcement learning with forgetting can be thought of as each agent in the population having an urn for each location in the string representation of their strategy profile. Each urn begins a simulation with one ball corresponding to each possible action for that location in the strategy profile string. So, in the system that produces the single embedding structure, there is one non-null signal in each of two dimensions, leading to four total signal combinations, and three possible actions in response to each signal. So an agent would have six total urns. In the first urn a null ball and a blue ball, in the second a null ball and a red ball, and in the third through sixth urns a triangle, diamond, and star ball in each urn. In a single timestep, an agent begins by drawing one ball from each of their urns at random with equal probability. These draws then determine the strategy profile that the agent employs for the entire timestep while interacting with every other agent in the population. For each individual that is interacted with, if an agent receives a non-zero payoff the urns that contributed to their actions are reinforced. Reinforcements can be thought of as adding additional balls of the type that was drawn to the urn that it was drawn from. For example, suppose two Girshites both employing the strategy profile  $\langle \bullet \blacklozenge \blacktriangle \star \rangle$  successfully coordinate on the Girshite greeting ( $\blacklozenge$ ). Then, they would each add payoff-many blue balls to their first urn, payoff-many red balls to their second urn, and payoff-many diamond balls to their third urn. Since the urns corresponding to what action they choose in response to the signals  $\bullet$ ,  $\blacklozenge$ , and  $\blacktriangle$  were not used for this interaction, they are not reinforced. Finally, at the end of a timestep, forgetting is implemented by multiplying the number of balls of each type by a forgetting factor  $0 < \phi < 1$ . For any ball type whose quantity falls below 1 in any urn, that quantity is raised to 1; this helps preserve the possibility of that action being chosen in the future.

To represent assortment, we adjusted agents payoff by multiplying it by  $H(h, x, i)$  where  $H(h, x, i)$  is defined as:

$$H(h, x, i) = \frac{N}{\sum_{j \in Y} [S(h, x, j)]} \times S(h, x, i)$$

where  $N$  is the total number of agents in the population and  $S(h, x, j)$  is:

$$S(h, x, j) = (2 * d)^h \quad \text{if profile } x \text{ entails broadcasting the same social signal as } j \text{ in } d\text{-many dimensions}$$

$$S(h, x, j) = 1 \quad \text{if } x \text{ does not entail broadcasting the same social signal as } j$$

We found that having agents select a single strategy profile that they employ for all interactions on a single timestep made this learning dynamic significantly more computationally tractable. We also suspect that it improves agents ability to learn in virtue of reducing the amount of noise in the system.

With these reinforcement learning dynamics, we ran simulations for a population of 500 agents where Lagashites were a  $0.33 + \beta$  proportion of the population, Girshites were a  $0.33 - \beta$  proportion of the population, Akkadians were a 0.34 proportion of the population, signal cost was -0.000125, and the forgetting factor,  $\phi$ , was 0.998. The coordination preferences were:

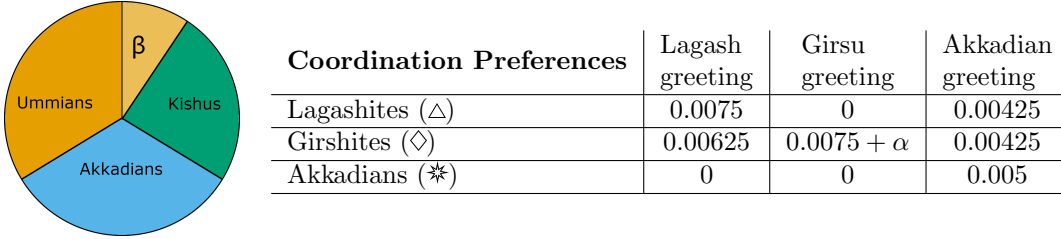


Table 1: Coordination preferences for the single embedding subsection.

This frequently produced the single embedding group structure. Note that, here again we see assortment making optimal outcomes more likely: